

Track 2.3: Text Cleaning in RAG Systems

Detailed Guidelines

Objective

The purpose of Text Cleaning is to convert raw extracted text from PDFs, DOCX files, websites, or scanned documents into a clean and standardized format before chunking and embedding generation.

A poorly cleaned document leads to:

- Irrelevant chunks
- Poor retrieval accuracy
- Increased hallucinations
- Wasted storage and computation

Why Text Cleaning is Important

Consider the following text extracted from a PDF:

Page 1

Research Paper on Artificial Intelligence

1. Introduction

Artificial Intelligence (AI) is transforming industries.

2. Literature Review

Problems:

- Page numbers
- Repeated headers
- Separator lines
- Extra blank spaces
- Formatting noise

If stored directly in a vector database, retrieval quality decreases significantly.

Text Cleaning Workflow

Raw Document
↓
Text Extraction
↓
Remove Noise
↓
Normalize Text
↓
Validate Output
↓
Chunking

Task 1: Remove Extra Spaces

Problem

Artificial Intelligence is transforming industries.

Solution

```
import re

text = re.sub(r'\s+', ' ', text)
```

Output

Artificial Intelligence is transforming industries.

Task 2: Remove Multiple Blank Lines

Problem

Introduction

Artificial Intelligence

Solution

```
text = re.sub(r'\n+', '\n', text)
```

Output

Introduction

Artificial Intelligence

Task 3: Remove Page Numbers

Problem

Page 1
Page 2
Page 3

Solution

```
text = re.sub(r'Page\s+\d+', '', text)
```

Task 4: Remove Headers and Footers

Problem

Every page contains:

```
XYZ University Research Report
```

Solution

```
text = text.replace(  
    "XYZ University Research Report",  
    ""  
)
```

For dynamic detection:

```
from collections import Counter
```

Identify frequently repeated lines and remove them.

Task 5: Remove Special Characters

Problem

@@@ Artificial Intelligence ###

Solution

```
text = re.sub(
    r'^a-zA-Z0-9\s.,!?:;()-]',
    '',
    text
)
```

Output

Artificial Intelligence

Task 6: Fix Broken Sentences

Problem

PDF extraction often gives:

```
Artificial Intelligence is  
transforming industries and  
creating new opportunities.
```

Solution

```
text = text.replace("\n", " ")
```

Output

```
Artificial Intelligence is transforming industries and creating new opportunities.
```

Task 7: Unicode Normalization

Problem

"AI" 'Machine Learning'

Different quotation marks may cause inconsistencies.

Solution

```
import unicodedata

text = unicodedata.normalize(
    "NFKD",
    text
)
```

Task 8: Remove URLs (Optional)

Problem

Visit <https://example.com>

Solution

```
text = re.sub(  
    r'http\S+',  
    '',  
    text  
)
```

Task 9: Remove Email Addresses (Optional)

Problem

Contact: abc@gmail.com

Solution

```
text = re.sub(  
    r'\S+@\S+',  
    '',  
    text  
)
```

Task 10: OCR Error Correction

Scanned PDFs often generate:

Machlne Learning
ArtificiaI Intelligence

Libraries:

```
pip install textblob  
pip install pyspellchecker
```

Example:

```
from spellchecker import SpellChecker  
  
spell = SpellChecker()
```

Note: Use carefully because technical terms may be incorrectly modified.

Task 11: Remove Boilerplate Text

Example

Copyright © 2026
All Rights Reserved

These lines are usually useless for retrieval.

Remove them using:

```
text = text.replace(  
    "All Rights Reserved",  
    ""  
)
```

Task 12: Preserve Important Structure

Do NOT remove:

- Section titles
- Headings
- Tables (if important)
- References (if needed)
- Technical terms
- Equations

Bad Example:

Chapter 3: Transformer Architecture

Removing this heading can reduce retrieval quality.

Recommended Libraries

Basic Cleaning

```
import re
import unicodedata
```

NLP-Based Cleaning

```
pip install nltk
pip install spacy
```

Libraries:

- NLTK
- SpaCy

Used for:

- Tokenization
 - Lemmatization
 - Stopword handling
-

Enterprise Document Processing

```
pip install unstructured
```

Useful for:

- PDFs
 - DOCX
 - PPTX
 - HTML
 - Emails
-

Sample Cleaning Function

```
import re
import unicodedata

def clean_text(text):

    text = unicodedata.normalize(
        "NFKD",
        text
    )

    text = re.sub(
        r'Page\s+\d+',
        '',
        text
    )

    text = re.sub(
        r'\s+',
        ' ',
        text
    )

    text = re.sub(
        r'http\S+',
        '',
        text
    )

    return text.strip()
```

Objective

Build a Text Cleaning Module for a RAG Pipeline.

Input

- PDF
- DOCX
- TXT

Tasks

1. Extract raw text.
2. Remove page numbers.
3. Remove headers/footers.
4. Remove unwanted symbols.
5. Normalize spaces.
6. Normalize Unicode characters.
7. Generate a cleaned text file.

Output

Input:
raw_document.txt

Output:
clean_document.txt